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Course: Data Analytics & Data Science

Assignment: 01

1. A database is an organized collection of information stored electronically so that it can be easily accessed, managed, and updated.

- Databases are essential because modern life generates huge amounts of data. Without databases, it would be nearly impossible to find, update or

Real-world example

- A hospital uses a database to store patient records, appointment schedules and medical histories
- An e-commerce site like Amazon uses a database to track products, customer orders and inventory.

2. A data warehouse is a large, centralized storage system designed to hold data from many different sources

- A regular database is built for fast day to day operation, a data warehouse is optimized for running complex queries and analyzing historical data.

- A company uses a data warehouse to get a single, reliable version of the truth.

3. A Data lake is a storage that holds raw data in its original format

- A data warehouse stores processed, cleaned data for specific purposes
- A business would prefer a data lake when it needs to store massive amounts of varied data

4. A data mart is smaller, focused version of a data warehouse.

- It contains data for just one team or department such as sales, finance or marketing.
- A data warehouse feeds data into data marts
- Department managers, business analysts and team leaders use data marts because they need fast access to relevant data without wading through irrelevant information.

5. Structured data.

- Highly organised and fits into tables with rows and columns

Semi-Structured data

- Has some structure but does not follow a rigid table format
- It often uses tags or markers to separate elements.

Unstructured data.

- No predefined format, it is raw and often text-heavy or media based.

6. Data modelling is the process of creating a visual diagram or blueprint that shows how data is connected and stored in a system.

- It helps designers avoid mistakes, ensures data is consistent and makes the system easier to understand and maintain.

Conceptual data model - High level showing main entities and their relationships. Used to discuss ideas with non-technical stakeholders.

Relational data model - Uses tables with keys to link data.

7. Data mining is the process of discovering hidden patterns, correlations or trends in large datasets using statistics and machine learning.

- Businesses use data mining to predict customer behaviour, detect fraud, improve marketing and increase sales.

8. A database management system is software that lets users create, read, update and delete data in a database while keeping it secure and efficient.

MySQL - Open-source relational DBMS used in small to medium websites.

PostgreSQL - Advanced open-source relational DBMS with strong data integrity.

Oracle Database - Commercial, high performance DBMS.

MongoDB - NoSQL document DBMS.

9. ETL stands for Extract, Transform, Load - a process that moves data from source systems to a data warehouse

Extract - pull raw data from sources like databases, CSV files or APIs

Transform - Clean, filters and merge and reformat the data

Load - Insert the transformed data into the target system like a data warehouse

Why is it important : ETL ensures that data analysts work with clean, consistent and reliable data.

- Without ETL, data reports might be wrong or misleading.

10. - CSV. Simple, row-based text format. Used when sharing small tables with non-technical people or opening in Excel

JSON - Stores nested, hierarchical data with key value pairs

Parquet : Columnar, compressed binary format. Used in big data analytics.

Avro - Row-based binary format with schema embedded and used when streaming data or evolving schemas over time.

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10. Artificial Intelligence is the field of computer science that creates machines capable of performing tasks that normally require human intelligence.

Narrow AI is designed for one specific task. Eg Netflix recommendation engine.

General AI is a hypothetical AI that can understand, learn and apply intelligence across many different tasks like a human. an Example is an AI that can cook, drive, write poetry and diagnose illness.

12. Machine learning is a subset of AI where computers learn from data without being explicitly programmed with rules.

Traditional programming : Rules + Data = Answer

Machine learning : Answer + Data = Rules.

Supervised learning

Unsupervised learning

Reinforcement learning

13. Deep learning is a ^{subset} ~~subject~~ of machine learning that uses many layers of artificial neural networks to learn very complex patterns.

Neural networks - Inspired by the human brain, they consist of layers of interconnected neurons. Each layer learns different features.

- They work by passing data forward, calculating errors and adjusting connection strengths backward in a process called back-propagation.

14. NLP is the branch of AI that helps computers understand, interpret and generate human language.

- Spell check and autocomplete in your phone keyboard
- Virtual assistants like Siri or Alexa
- Email spam filters that read the message content

15. LLM is a type of ~~data~~^{deep} learning model trained on massive amounts of text to understand and generate human-like language.

Standard ML models are small and task-specific.

LLM are huge and can perform many language tasks without retraining.

e.g. ChatGPT and Claude.

16. Generative AI creates new content from patterns learned from training data.

- It learns the statistical pattern in data and then samples from those patterns to produce new original examples that look realistic.

- ChatGPT
- Dall-E 3
- Midjourney

17. • A training dataset is the collection of examples used to teach an AI model how to perform a task

- The AI learns only from what it sees
- If a facial recognition system is trained mostly on light-skinned faces it will perform poorly on darker-skinned faces leading to false arrests or discrimination
- If a hiring AI is trained on resumes or past male hires it may unfairly reject females.

18. Computer vision is a field of AI that enables machines to interpret and understand visual information from images or videos

- They convert images into numbers then use neural networks to detect edges, shapes, objects and patterns layer by layer until they can recognise a face, car or cat.
- Self driving cars
- Medical imaging.

19. - An API is a set of rules that allows one software application to talk to another

- ~~How~~ When you give a waiter in a restaurant your order and the waiter takes it to the kitchen and brings back your food
- APIs let different systems work together without needing to know each other's internal details
- They save time, increase security and allow companies to build powerful services.

20. An API is a unique code that identifies and authenticates a developer or application calling an API.

- It is used to control access, track usage and prevent abuse
- Store API keys in environment variables or secure secret managers
- Attackers can steal the key, run up huge bills, steal data or delete resources

21. REST is an architectural style of designing APIs that use standard web technologies

GET - Retrieve data

POST - Create new data

PUT - Update existing data completely

DELETE - Remove data

22. JSON is a lightweight text format for storing and exchanging data that is easy for humans to read and for machines to parse

- It is language-independent, compact and works well with web technologies

E.g. "Student"

"name" Mariam

"age" 22

"courses" "python"

23 Prompt engineering is the practice of designing and refining inputs to get the best possible output from an AI language model

- A poorly written prompt leads to vague, wrong or useless answers

- Good prompts saves time and improves results

"Tell me about data"

24.- A map ^{function} ~~python~~ applies a given operation to every item in a list and returns a new list of result

Python

```
number = [1, 2, 3, 4, 5]
```

```
Squared = list(map(lambda x: x**2, numbers))
```

```
print(Squared) # output: [1, 4, 9, 16, 25]
```

25 ~~##~~ - Cloud computing is delivering computing services over the internet on a pay as you go basis

- * IaaS : virtual servers and storage

- * PaaS : platform to deploy apps without managing hardware

- * SaaS : Ready to use software over the internet.

26.- A Version Control System tracks changes to files overtime so you can go back to earlier versions and collaborate without overwriting work

- Git is the most popular version control system used by data analysts and developers to manage code and notebooks

- A snapshot of files at a point in time, with a message describing the ^{change} documents
- A separate line of development. You can experiment on a branch without affecting the main version

27. - A Jupyter notebook is an interactive, web-based tool that allows you to write and run code, see output, add text, images and visualisations in a single document

- The most common language: Python
- It combines code, results and explanations in one place, making it perfect for exploratory analysis, teaching and reproducible research

28. - Python is a high level, general purpose programming language known for its simple, readable syntax

- It is popular because it is easy to learn and write
- Strong community ~~and~~ industry support
- Huge ecosystem of libraries for math, stats, ML and visualisation

Python libraries: Pandas
Numpy
Matplotlib
Scikit learn.

29. - Data privacy is the right of individuals to control how their personal information is collected, used and shared

- It is important to prevent and identify theft, discrimination, financial loss or trust.
- GDPR is a strict European Union law that gives EU citizens rights over their data
- It applies to any company that processes data of people in the EU even if the company is outside Europe

30. - Data ethics is the branch of ethics that studies the moral issues related to data collection, analysis and use

- Facial recognition without consent
- Selling user data without permission
- AI used to deny loans based on zip code

31. - Data governance is the overall management of data availability, usability, integrity and security in an organisation
key components:

- Data quality
- Data security and privacy
- Data stewardship
- Compliance
- Data lifecycle management

32. - Bias in AI means model produces systematically prejudiced results due to flawed assumptions, skewed training data or poor design

- Amazon's recruiting AI penalised resumes containing the word women

Consequences

- Unfairly excluding qualified candidates
- Reinforcing existing societal inequalities
- Legal liability and reputation damage

33.

	Data Analyst	Data Scientist	Data Engineer	AI Engineer
Role	Answer business questions with historical data	Build predictive models and fight insights	Build and maintain data pipelines	Deploy and scale AI/ML models into production
Skills	SQL, Statistics data visualisation	ML, Statistics programming	ETL, database cloud, data warehousing	ML, Software engineering, APIs
Tools	Excel, Table Powerpoint	Python, R Scikit learning	Spark, Airflow AWS	Docker Kubernetes

34. Business Intelligence is the process of collecting analysing and presenting business data to help managers make informed decisions

- Power BI - powerful data visualization and real-time analytics for large datasets
- Tableau - powerful data visualisation and real-time analytics for large datasets

They help turn raw data into easy to understand charts and dashboards so leaders can spot trends

35 - I found LLMs and Generative AI the most interesting.
- I did not realise that the same model can write poetry summarise legal documents, debug codes and have a conversation without being specially trained for each task

36. * Deep learning and neural networks were hardest for me, especially how backpropagation works mathematically and why many layers help. I used YouTube videos, Google and explained it to a friend

37. * In my future career I think AI will handle repetitive data cleaning and reporting, freeing me to focus on asking better questions and interpreting results